

PS 1

b) (A5) - V

$$A_{V-R} = 0.4$$

$$(A_{V-R})_0 = 0.13$$

$$M_V = 5.0$$

$$V: \lambda_V = 5500 \text{ \AA}$$

$$R: \lambda_R = 6500 \text{ \AA}$$

$$A_\lambda = \frac{1}{\lambda}$$

$$\bullet E(V-R) = A_{V-R} - (A_{V-R})_0 = 0.4 - 0.13 \\ = A_V - A_R = 0.27$$

$$A_V \left(1 - \frac{A_R}{A_V}\right) = 0.27$$

$$\bullet A_V \left(1 - \frac{\lambda_V}{\lambda_R}\right) = A_V \left(1 - \frac{5500}{6500}\right) = 0.27$$

$$A_V \left(\frac{2}{13}\right) = 0.27$$

$$\boxed{\begin{array}{l} A_V = 1.75 \\ A_R = 1.48 \end{array}}$$

$$\textcircled{c} M_V = +1.8$$

$$m_V - M_V = 5 \log(D) - 5 + A_V$$

$$5.0 - 1.8 = 3.2 = 5 \log(D) - 5 + A_V$$

$$\frac{3.2 - 1.75}{5} + 1 = \log_{10}(D)$$

$$10^{(\quad)} = D$$

with out A_V

$$A \ 3 \mid m_V = 14.75 \quad \text{period: } T_0 = 224 \text{ hrs}$$

$$\log\left(\frac{L_v}{L_{v0}}\right) = 0.9 \log(P) + 2.75$$

$$I_v = \frac{L_v}{4\pi D^2}$$

$$I_{v0} = \frac{L_{v0}}{4\pi D_0^2}$$

$$\frac{I_v}{I_{v0}} = \frac{L_v}{L_{v0}} \frac{D_0^2}{D^2} = (100)^{\frac{1}{5}(M_0 - M)}$$

$$\ln\left(\frac{I_v}{I_{v0}}\right) = \frac{2}{5}(M_0 - M_v)$$

$$\frac{2}{5}(M_0 - M_v) = \ln\left(\frac{L_v}{L_{v0}}\right) + \ln\left(\frac{D_0^2}{D^2}\right)$$

$$\frac{2}{5}(M_0 - M_v) = 0.9 \log_{10}(P) + 2.75 + \ln\left(\frac{D_0^2}{D^2}\right)$$

$$\frac{2}{5}(4.83 - 14.75) = 0.9 \log_{10}(P) + 2.75$$

$$= 2 \log_{10}\left(\frac{10}{D}\right)$$

$$D = 10 \text{ (H)}$$

• Give in terms of

$$D_0 = 10$$



